ClinicOS

Clinic Management & Patient Experience Platform

Product Requirements Document (PRD)

Version 1.0 | Confidential

Date: 26 February 2026

1. Project Overview & Goals

1.1 Executive Summary

ClinicOS is an all-in-one clinic management platform designed to digitize and streamline the complete patient lifecycle — from appointment booking and queue management to billing, history tracking, and automated communication. It eliminates paper-based processes, reduces wait times, and gives clinics a powerful yet simple operating system.

1.2 Problem Statement

- Patients waste hours waiting without any visibility into queue status
- Doctors lack instant access to previous visit records, medicines, and complaints
- Clinic staff manually manage queues, leading to errors and delays
- Bills are handed over on paper with no digital trail
- No automated communication means patients miss their turn or arrive too early/late

1.3 Solution

ClinicOS provides a role-based web platform for Patients, Doctors, and Clinic Staff — each with a tailored interface — powered by real-time queue management, automated WhatsApp/email notifications, complete visit history, and live clinic analytics.

1.4 Strategic Goals

- Reduce average patient wait time by 40% within 3 months of deployment
- Eliminate paper-based patient registration and billing
- Enable doctors to access full patient history in under 5 seconds
- Automate 100% of routine patient communications via WhatsApp and email
- Give clinic owners real-time financial and operational visibility

2. Target Audience

2.1 Primary Users

User Role	Profile	Key Needs
Patient	Age 18–70, smartphone user, visits clinic 1–6x per year	Book appointments, check queue, receive notifications, view bills & history
Doctor	Practicing physician or specialist, tech-moderate, time-constrained	View patient queue, access full history, manage consultation flow, pause queue
Clinic Staff / Receptionist	Front-desk admin, manages daily operations	Register patients, manage queue, generate bills, monitor clinic stats
Clinic Owner / Admin	Business owner overseeing one or more clinics	Revenue reports, staff management, analytics, clinic-wide settings

2.2 Secondary Users

- Pharmacists affiliated with the clinic needing prescription access
- Lab technicians receiving test referrals from doctors
- Insurance verifiers requiring billing summaries

3. Complete Website Page Structure

3.1 Public Pages

- / — Landing Page (hero, features, CTA, testimonials, pricing)
- /login — Unified Login with role detection
- /signup — Registration with role selection (Patient / Doctor / Staff)

- /forgot-password — Password reset flow
- /services — Public clinic services listing page
- /about — Clinic information and doctor profiles

3.2 Patient Portal (/patient/)

- dashboard — Overview: next appointment, token status, recent visits
- book — Book appointment: select service, doctor, date/time slot
- queue — Live queue tracker: current token, position, estimated wait
- history — Full visit history: complaints, medicines, bills per visit
- bills — Bill listing and detail view, download as PDF
- profile — Personal details, phone number, preferences
- notifications — Notification log (WhatsApp/email history)

3.3 Doctor Portal (/doctor/)

- dashboard — Current patient info, quick actions, daily summary
- queue — Full queue view: waiting list, token numbers, estimated times
- patient/:id — Full patient profile: history, medicines, complaints, bills
- consultation — Active consultation view with notes, prescription pad
- history — Past consultation records by patient or date
- settings — Availability schedule, lunch break defaults

3.4 Clinic Staff Portal (/staff/)

- dashboard — Daily overview: patients added, queue status, revenue
- register — Patient entry: phone number → auto-fill or register new
- queue — Full queue management: reorder, remove, pause, emergency
- billing — Generate bill: services, medicines, taxes, send via SMS/email
- stats — Clinic statistics: patients today, revenue, avg wait, complaints
- patients — Patient directory with search and filter

3.5 Admin Portal (/admin/)

- overview — Clinic-wide KPIs and multi-doctor summary
- staff — Manage staff accounts and permissions
- doctors — Manage doctor profiles and schedules
- services — Add/edit/delete clinic services and pricing
- reports — Exportable monthly/weekly reports (revenue, patients, feedback)
- settings — Clinic name, logo, WhatsApp API config, email config, SMS templates

3.6 Shared / System Pages

- /404, /500 — Error pages
- /terms, /privacy — Legal pages
- /help — Help center / FAQ

4. Core Features with Priority Levels

Priority Scale: P1 = Must-Have (MVP) | P2 = Important (v1.1) | P3 = Nice-to-Have (v2)

4.1 Feature Registry

Feature	Description	Priority	Module
Smart Patient Entry	Enter phone → fetch existing patient or register new. Zero redundant data entry.	**P1**	All
Live Queue Management	Real-time token system with position, estimated wait, and live updates.	**P1**	Staff / Doctor
WhatsApp Automation	Auto-send messages on patient add, 2-patient warning, and turn arrival.	**P1**	System
Email Automation	Parallel email notifications for every WhatsApp message trigger.	**P1**	System
Doctor Memory (History)	Full per-patient history: visits, medicines, complaints, bills, dates.	**P1**	Doctor
One-Tap Queue Controls	Pause queue (lunch), send to lab + return, emergency priority insertion.	**P1**	Doctor / Staff
Digital Billing	Generate itemized bill and send via WhatsApp/SMS/email automatically.	**P1**	Staff
Clinic Stats Dashboard	Real-time: patients today, revenue, avg wait, top complaints.	**P1**	Admin / Staff

Feature	Description	Priority	Module
Appointment Booking	Patient books slot online; staff receives notification.	P2	Patient
Patient Portal	Patient self-service: queue status, history, bills, profile.	P2	Patient
PDF Bill Download	Patient can download their bill as PDF anytime.	P2	Patient / Staff
Doctor Availability Calendar	Doctors set available hours; patients see real slots.	P2	Doctor
Multi-Doctor Clinic Support	Separate queues and dashboards per doctor within one clinic.	P2	Admin
Feedback & Rating	Patient rates visit after completion; shown in admin stats.	P3	Patient
Prescription PDF	Generate and send prescription as PDF via WhatsApp.	P3	Doctor
Lab Test Tracking	Track which tests were ordered, results status, and return-to-queue.	P3	Doctor / Staff
Inventory / Medicine Stock	Basic medicine inventory linked to prescriptions.	P3	Admin
Multi-Clinic / Franchise	One admin managing multiple clinic branches.	P3	Admin

5. Feature Specifications

Feature 1 — Smart Patient Entry

How it works

1. Staff opens Register screen and types patient's phone number.
2. System queries database in real-time (debounced, 300ms).

3a. Existing patient → auto-fills name, age, address, full visit history. Staff confirms and adds to queue.

3b. New patient → form appears with required fields (name, age, gender, address, DOB). On submit, patient is created and added to queue.

4. System assigns the next available token number and triggers WhatsApp notification.

Acceptance Criteria

- Phone number lookup must return results within 500ms.
- New patient registration must complete in under 3 form fields minimum (name, phone, age).
- Token is assigned immediately upon queue entry — no manual step.

Feature 2 — Live Queue Management

Queue States & Display

Each patient in queue has a status: Waiting | With Doctor | Paused (Lab) | Done | Emergency
Queue screen shows: Token # | Patient Name | Status | Estimated Wait Time
Estimated wait = avg consultation time × position in queue (configurable per doctor)
Queue is visible to: Doctor (their portal), Staff (management view), Patient (their own position only)

One-Tap Controls (Feature 5 Integration)

- Lunch Pause — freezes queue timer; all patient ETAs pause; WhatsApp sent to next 3 patients
- Send to Lab — removes patient from active consultation, holds their token; returns when staff taps 'Back from Lab'
- Emergency Insert — places patient at position #1 instantly, bumping others down by one; notification sent

Feature 3 — WhatsApp & Email Automation

Trigger	WhatsApp Message Template	Timing
Patient Added to Queue	*Hi [Name]! You have been added to the queue. Your token is #[Token]. Current wait time is approx. [X] minutes.*	Immediately on entry
2 Patients Ahead	*Hi [Name], only 2 patients ahead of you. Please make your way to the clinic soon!*	Auto-triggered at 2 ahead
Your Turn	*Hi [Name], it's your turn! Please come to the doctor's room immediately. Token #[Token].*	When doctor calls next
Queue Paused (Lunch)	*Hi [Name], the doctor is on a short break. Queue resumes in ~[X] min. Your wait time is updated.*	On lunch pause

Trigger	WhatsApp Message Template	Timing
Bill Ready	*Hi [Name], your bill of Rs.[Amount] is ready. Download: [Link]. Thank you for visiting!*	After billing
Appointment Confirmed	*Hi [Name], your appointment with Dr. [Name] is confirmed for [Date] at [Time]. Token will be assigned on arrival.*	On booking

Feature 4 — Doctor Memory (Patient History)

Data Captured Per Visit

Visit Date & Time | Doctor Name | Clinic Branch
Chief Complaint (text + tags) | Diagnosis
Medicines Prescribed (name, dosage, duration, instructions)
Lab Tests Ordered | Lab Results (if uploaded)
Consultation Notes (doctor's private notes)
Bill Amount & Breakdown | Payment Status
Follow-up Date (if set)

The Doctor's consultation screen shows a right-side panel with the patient's last 5 visits collapsed, expandable with one click. A global 'Full History' button opens a timeline view of all visits.

Feature 6 — Digital Billing

- Staff selects services rendered, medicines dispensed, and any additional charges.
- System auto-calculates subtotal, taxes (GST configurable), and total.
- Bill is generated as a PDF with clinic letterhead, doctor name, and QR code.
- Sent automatically via WhatsApp and email on bill generation.
- Patient can download bill anytime from their portal.
- Bill status tracked: Generated | Paid | Partially Paid | Pending.

Feature 7 — Clinic Stats Dashboard

Key Metrics Displayed

Today's Patients: Total visited | Currently waiting | Completed
Revenue: Today | This Week | This Month | By Doctor
Average Wait Time: Today | 7-day trend chart
Top Complaints: Most frequent diagnoses/complaints (bar chart)
No-Show Rate: Appointment booked but not attended
Doctor Utilization: Consultations per doctor per day

Queue Health: Current wait time vs clinic target

6. User Journey Mapping

Journey 1 — Walk-in Patient (Most Common)

#	Step	System Action	User Experience
1	**Patient arrives at clinic**	—	Patient walks up to reception
2	**Staff enters phone number**	Lookup in DB → return patient data or new form	Staff sees patient auto-populated in 1 sec
3	**Patient added to queue**	Token assigned, WhatsApp sent with token + ETA	Patient receives WhatsApp with token # and wait time
4	**Patient waits**	Queue updates live as other patients are seen	Patient can check their queue position on their phone
5	**2 patients ahead**	System auto-sends WhatsApp reminder	Patient gets 'Come soon' message
6	**Patient's turn**	Doctor taps 'Next Patient'; WhatsApp sent immediately	Patient receives 'It's your turn' alert
7	**Consultation happens**	Doctor views history, adds notes, prescribes medicine	Doctor has full context; patient gets personalized care
8	**Bill generated**	Staff creates bill; sent via WhatsApp + email as PDF	Patient receives digital bill on phone
9	**Patient leaves**	Visit record saved; history updated	Complete record stored for next visit

Journey 2 — Online Appointment Booking

- Patient visits ClinicOS website → clicks 'Book Appointment'
- Selects service category → selects doctor → selects available date and time slot
- Enters phone number → existing patient auto-fills, new patient registers
- Confirms booking → receives WhatsApp and email confirmation with appointment details
- On appointment day → arrives at clinic → staff checks them in → token assigned → normal queue flow begins

Journey 3 — Emergency Patient

- Emergency patient arrives; staff taps 'Emergency' button on queue screen

- Patient is instantly placed at Position #1 regardless of existing queue
- All other patients in queue receive automatic delay notification
- Doctor is notified immediately; consultation begins within seconds

Journey 4 — Doctor Lunch Break

- Doctor taps 'Pause Queue' (one tap on dashboard)
- All patient ETAs are paused and recalculated with break duration
- Next 3 patients in queue receive 'Queue paused — break in progress' WhatsApp
- Doctor taps 'Resume Queue' when back → ETAs resume → patients notified

7. Success Metrics (KPIs)

7.1 Operational Metrics

Metric	Baseline (Before ClinicOS)	Target (3 months)	Target (6 months)
Average Patient Wait Time	45–90 minutes	< 40 minutes	< 25 minutes
Patient Registration Time	5–8 minutes (paper)	< 45 seconds	< 20 seconds
No-Show Rate	~25% (untracked)	< 15%	< 10%
Bill Generation Time	10–15 minutes	< 2 minutes	< 1 minute
WhatsApp Delivery Rate	0% (none)	> 95%	> 98%
Repeat Patient Recognition	Manual / memory	100% automated	100% automated
Daily Patient Capacity	Baseline clinic max	+15% throughput	+30% throughput

7.2 Business Metrics

- Month-over-Month Revenue Growth tracked in admin dashboard
- Clinic Retention Rate: % of clinics renewing subscription after 3 months (target: > 80%)
- NPS Score from Patient Feedback (target: > 50 within 6 months)
- Feature Adoption Rate: % of clinics using WhatsApp automation (target: > 90%)

7.3 Technical Metrics

- API Response Time: < 300ms for all queue and patient lookup endpoints
- System Uptime: 99.5% monthly (allow max 3.6 hours downtime/month)
- WhatsApp Message Delivery Time: < 10 seconds from trigger event
- Page Load Time: < 2 seconds on 4G mobile connection
- Concurrent Users: Support 200 simultaneous staff/doctor sessions per clinic deployment

8. Recommended Technology Stack

8.1 Frontend

- Framework: React.js (Next.js for SSR + SEO on public pages)
- Styling: Tailwind CSS
- State Management: Zustand or Redux Toolkit
- Real-time: Socket.IO client (for live queue updates)
- Charts: Recharts or Chart.js (clinic stats dashboard)

8.2 Backend

- Runtime: Node.js with Express.js or Fastify
- Database: PostgreSQL (primary) + Redis (queue state, sessions, pub-sub)
- Real-time Server: Socket.IO (queue broadcasts)
- Authentication: JWT with refresh tokens; role-based access control (RBAC)
- File Storage: AWS S3 or Cloudflare R2 (PDF bills, prescriptions, lab reports)

8.3 Integrations

- WhatsApp: Meta WhatsApp Business API or Twilio WhatsApp Sandbox
- Email: SendGrid or AWS SES
- SMS Fallback: Twilio SMS or MSG91 (India)
- PDF Generation: Puppeteer or PDFKit
- Payments (v2): Razorpay or Stripe for online payment before appointment

8.4 Infrastructure

- Hosting: AWS / GCP / Railway.app (for MVP speed)
- CI/CD: GitHub Actions
- Monitoring: Sentry (errors) + Datadog or BetterStack (uptime + logs)
- Security: HTTPS enforced, data encrypted at rest (AES-256), GDPR/DPDP compliant

9. Non-Functional Requirements

9.1 Security

- All patient data encrypted at rest (AES-256) and in transit (TLS 1.3)
- Role-based access: patients cannot see other patients; doctors see only their queue
- Phone OTP verification required for patient portal login
- Admin actions logged with timestamp and user ID for audit trail
- HIPAA-adjacent compliance: no PHI shared with third parties except messaging APIs

9.2 Accessibility

- WCAG 2.1 AA compliance for all public-facing pages
- Mobile-first responsive design (primary usage is mobile in clinic settings)
- Support for low-bandwidth environments (3G-optimized API payloads)

9.3 Scalability

- Multi-tenant architecture: each clinic is an isolated tenant with its own data
- Horizontal scaling: stateless API servers behind load balancer
- Database connection pooling with PgBouncer

9.4 Data Retention

- Patient visit records retained for minimum 7 years (medical standard)
- Billing records retained for minimum 5 years (tax compliance)
- Soft delete only — no hard deletion of patient or billing data

10. MVP Scope & Phased Roadmap

Phase 1 — MVP (Weeks 1–8)

Included in MVP

✓ Patient entry by phone number (new + returning)
✓ Token-based queue system with real-time position tracking
✓ WhatsApp notifications: added to queue, 2-ahead warning, your turn
✓ Doctor consultation screen with patient history
✓ One-tap controls: lunch pause, emergency insert, send to lab

✓ Basic billing: generate, send via WhatsApp
✓ Clinic Stats Dashboard: daily patients, revenue, wait time
✓ Role-based login: Patient, Doctor, Staff, Admin

Phase 2 — v1.1 (Weeks 9–14)

Added in Phase 2
+ Online appointment booking with slot management
+ Patient self-service portal (queue status, history, bills)
+ PDF bill download and prescription PDF generation
+ Email notifications parallel to WhatsApp
+ Multi-doctor support within one clinic
+ Doctor availability calendar and schedule management
+ Advanced clinic reports: weekly/monthly export

Phase 3 — v2.0 (Weeks 15+)

- Patient feedback and rating system
- Lab test tracking with result upload
- Online payment before appointment (Razorpay)
- Multi-clinic / franchise admin panel
- Medicine inventory management
- Telemedicine / video consultation integration
- AI-powered: diagnosis suggestions, drug interaction alerts

11. Assumptions & Risks

11.1 Assumptions

- Clinics have at least one smartphone-capable staff member at reception
- WhatsApp Business API access is obtained before launch (Meta approval required)
- Patients have WhatsApp installed on their registered phone number
- Clinic WiFi is available for staff and doctor devices
- Initial deployment targets single-doctor clinics in India (expandable)

11.2 Risks & Mitigations

Risk	Severity	Mitigation
WhatsApp API approval delays	**High**	Use Twilio sandbox for MVP; apply for Meta API in parallel
Doctors resist technology adoption	**Medium**	Provide a tablet-optimized, single-screen UI with training videos
Real-time queue sync failures	**High**	Implement optimistic UI + server reconciliation; offline fallback mode
Patient privacy data breach	**Critical**	End-to-end encryption, regular security audits, minimal PII storage
Queue timer inaccuracy	**Medium**	Allow doctor to set custom avg. consultation time; auto-adjust with ML

11.3 Open Questions for Stakeholders

- Will ClinicOS be SaaS (subscription) or sold as a one-time license per clinic?
- Which WhatsApp Business API provider is preferred (Meta direct vs Twilio vs 360dialog)?
- Should patient login require OTP only, or also support password-based login?
- Is multi-language support required at launch (Hindi, Gujarati)?
- What is the target clinic size for MVP: solo doctor, 2–5 doctors, or larger?